

AN2DL - Second Challenge Report

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1 Introduction

The problem being tackled in this project is related to classifying a *molecular subtype* based on low-magnification histopathology image patches. Every instance needs to be classified into one of these four categories: **Luminal A**, **Luminal B**, **HER2(+)**, and **Triple Negative**. The metric used during evaluation in this problem is **macro F1-score**.

Our aims were:

- constructing an accurate classifier which is able to generalize on a secret test subset used in leaderboard evaluation;
- make use of the given auxiliary masks if they are helpful in improving performance rather than creating a negative effect on generalization;
- achieve a stable performance by tracking experiments and using variance reduction methods.

Our last model is based on: (i) transfer learning with a pre-trained CNN architecture (**EfficientNet-B0**), (ii) aggressive data augmentation methods suitable for histopathology, and (iii) ensemble methods with **Test-Time Augmentation** and a **multi-seed** strategy. The leaderboard performance using a 3-seed ensemble with Test-Time Augmentation was **0.3784 macro F1**.

2 Problem Analysis

Dataset characteristics

The dataset consists of histopathology image patches paired with auxiliary binary masks. It includes **691** image/mask pairs for training and **477** image/mask pairs for testing (test labels are

not provided). Ground-truth labels for the training set are given in `train_labels.csv`. Since images have varying spatial resolutions, a consistent preprocessing pipeline (resizing and normalization [1]) is required.

Main challenges

1. **Limited training data:** only 691 labeled samples for a 4-class problem increases the risk of overfitting.
2. **High intra-class variability and inter-class similarity:** molecular subtypes can share similar morphology at low magnification.
3. **Domain variability:** stain and illumination differences can reduce generalization to the hidden test subset.
4. **Macro F1-score sensitivity:** the metric rewards balanced per-class performance, so improvements must not come at the expense of harder/minority classes.

Initial assumptions

We assumed transfer learning would be necessary due to the dataset size, and that the masks could be useful mainly for localization (ROI focusing) rather than naive pixel-wise masking.

3 Method

Preprocessing

Images are first translated into RGB matrices, then resized to a fixed resolution, which in our case is 224×224 . Finally, we apply **ImageNet normalization**, since the resulting inputs match the expectations of the pre-trained backbones used in our approach.

Mask usage

We pursued two main strategies for deploying the auxiliary masks:

- **Mask multiplication** ($I \leftarrow I \odot M$): the mask is binarized and broadcasted along channels. This suppresses background but can also remove contextual cues and may introduce distribution artifacts, such as large constant regions after normalization.
- **Mask-informed ROI**: after identifying the positive mask region, we compute the bounding box and expand it by a margin. The image is then cropped to this ROI and resized. If the mask is empty or too small, we fall back to using the full image. This leverages the mask for localization while maintaining useful context.

Data augmentation

Training-time augmentation is designed to introduce invariance to orientation, scale, and stain variation:

- **RandomResizedCrop** to enforce robustness to scale and translation;
- random **horizontal** and **vertical flips** (each with probability 0.5);
- **probabilistic rotations** with small angles;
- **ColorJitter** for stain/illumination robustness.

The best leaderboard settings of ColorJitter were:

- $p = 0.7$,
- brightness = 0.1,
- contrast = 0.1,
- saturation = 0.1,
- hue = 0.02.

Validation and test use only resize + normalization, without stochastic augmentation.

Architecture of the model

We use the pre-trained **EfficientNet-B0** [3] trained on ImageNet (via `timm`). We replace the original head with a 4-class classification layer. This provides a good balance between representational power and computational cost on Colab GPUs.

Optimisation

We train using **cross-entropy loss** and **AdamW** [2], while a **cosine annealing** scheduler stabilizes convergence. We track all runs using **Weights & Biases** and save the best checkpoint of each run (by validation macro F1) to enable reproducible inference.

Inference-time strategy

We employ two inference-time techniques:

- **Test-Time Augmentation (TTA)**: we consider the original image, H-flip, V-flip, and HV-flip, and average the predicted probabilities over these deterministic transforms.
- **Multi-seed ensemble**: we train multiple models with different random seeds and average their predicted probabilities, with each model already using TTA. This reduces variance due to randomness in training.

4 Experiments

Experimental protocol

We iterated in a staged fashion:

1. **Transfer learning baseline**: pre-trained backbone with standard augmentation.
2. **Mask strategy evaluation**: compare different ways of exploiting the auxiliary masks.
3. **Optimization via sweeps**: tune hyperparameters of both augmentation and the optimizer using Weights & Biases sweeps.
4. **Inference improvements**: add Test-Time Augmentation (TTA) and ensembling.

Ablation studies

Baseline. iterated baseline starting **resnet18**, EfficientNet-B0 gave us the least complex model with high enough performance hence was kept as baseline.

Masks. We compared three alternatives: (i) mask multiplication, (ii) ROI-based crops obtained from the mask bounding box, and (iii) ignoring masks. While naive pixel-wise masking was found to weaken generalization performance in our setting, ROI cropping using bounding boxes provided a clear improvement.

Label smoothing. We experimented with label smoothing using $\epsilon = 0.05$ and $\epsilon = 0.1$. Although it slightly increased validation macro F1 in some runs, it consistently decreased public leaderboard performance, therefore it was not used in the final submission.

ColorJitter tuning. We tried several candidate configurations for ColorJitter. Conservative settings performed poorly on the leaderboard. The configuration reported in the Method section yielded the best public improvement.

Hyperparameter search

We used W&B sweeps to explore learning rate, weight decay, and augmentation-related parameters (e.g., ROI margin and rotation probability). Sweeps were used to identify a strong region of hyperparameters to explore, and final results were produced by combining the best checkpoints with inference-time strategies (TTA and ensembling).

5 Results

Table 1 summarizes the public leaderboard progression for our main improvements.

Table 1: Public leaderboard macro-F1 for key variants.

Variant	Public macro-F1
Single model + TTA	0.3548
Single model + TTA + ColorJitter	0.3557
3-seed ensemble + TTA	0.3784

The strongest single improvement was obtained by adding **ensembling** on top of TTA, suggesting that variance due to training randomness is significant in this dataset. The ColorJitter augmentation provided a smaller but consistent gain, likely improving robustness to stain/illumination variability.

6 Discussion

Generalization-oriented techniques were more effective than aggressive architectural changes in this setting. In particular:

- **TTA** offered an immediate performance gain by enforcing orientation invariance at inference.
- **Ensembling** provided the largest improvement, reducing variance and improving robustness on the hidden evaluation subset.
- **Mask usage requires care:** direct masking wasn’t very useful; using masks instead for ROI localization was more stable.
- **Validation is not always predictive:** label smoothing only improved validation but didn’t generalize well to leaderboard test data.

A limitation of our workflow is that we only used EfficientNet-B0 in ensembling due to tuning time complexity. Future Work might explore tuning other models and combining them in the ensemble for better generalization.

7 Conclusions

We addressed histopathology molecular subtype classification using an ensemble of 3 EfficientNet-B0 models, utilizing mask-informed ROI cropping, strong data augmentation and inference-time variance reduction techniques. Our best public leaderboard score was **0.3784 macro F1**.

References

- [1] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. Imagenet: A large-scale

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- [2] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)*, 2019.
- [3] M. Tan and Q. V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2019.